

Report R25 — the algebraic treatment

A reading path and glossary for a physicist who learned quantum mechanics the Hilbert-space way

Tier 1 analysis

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Abstract

Reports R18–R24 of this project are written in a language — commutants, von Neumann algebras, multiplicative domains, noiseless subsystems — that a standard physics curriculum does not teach. This document is the missing prerequisite, assembled for thesis use. §1 states the five conceptual moves that separate the algebraic formulation from the one taught in a physics degree, since knowing *which* habits to unlearn is most of the difficulty. §2 gives a reading path with an explicit minimum: three items suffice to work in this area, and the rest is reference material. §3 states the three structure theorems everything else rests on, in a form a physicist can use. §4 is a glossary of some forty terms, each with a definition, an intuition, and the reference to cite. §5 lists the notational collisions in this literature that are known to cause errors — including three distinct uses of “strong versus weak”, two of which this project uses. §6 maps the vocabulary onto where each piece is used in R18–R24, so a thesis chapter can cite in the right direction.

Every journal reference marked ✓ in §6 was verified against the publisher or arXiv while this document was written. Textbooks are marked †: their bibliographic data is standard and was not re-verified, so check the edition before citing a page number.

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1 Five moves a physics degree does not teach

The algebraic formulation is not a different theory. It is the same quantum mechanics with the logical order reversed, and the reversal buys structure theorems that are invisible in the usual presentation. Five specific habits have to be given up.

1. Observables are primary; states are functionals on them. The usual order is: fix a Hilbert space $\mathcal{H}$; states are unit vectors or density matrices; observables are self-adjoint operators. The algebraic order is: fix a C^* -algebra $\mathcal{A}$ of observables, and a *state* is a positive linear functional $\omega : \mathcal{A} \rightarrow \mathbb{C}$ with $\omega(\mathbb{1}) = 1$. The Hilbert space is then *derived* from the algebra and the state, by the GNS construction. For finite-dimensional problems this reversal changes nothing computationally — but it is what makes “the algebra of observables that survive” a well-posed object, and that object is the subject of R22 and R24.

2. A subsystem need not be a tensor factor of the whole. This is the most important move, and the one with no counterpart in an undergraduate course. In a physics degree, “subsystem” means $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ globally. The structure theorem (§3.2) says the general situation is

$$\mathcal{H} \cong \bigoplus_k \mathcal{H}_{n_k} \otimes \mathcal{H}_{m_k},$$

a tensor product *inside each block* of a direct sum. A subsystem in the algebraic sense is one of the $\mathcal{H}_{n_k}$ factors: a genuine quantum degree of freedom, living inside a superselection block, which does not factor out of $\mathcal{H}$ globally. Every decoherence-free subsystem, every noiseless subsystem and every subsystem code [18] is of this form. If only one of the two readings of “subsystem” is in your head, this literature is confusing at every step.

3. A symmetry is an algebra, not a group. The usual object is a group G acting by unitaries U_g . The algebraic object is the *commutant* $\{U_g\}' = \{A : [A, U_g] = 0 \ \forall g\}$, and it is the commutant that the structure theorems see. Two consequences. A group is only one way of generating a commutant, and the interesting commutants in fragmentation are generated by no group at all — that is precisely Moudgalya and Motrunich’s point [25]. And degeneracy is a statement about the commutant being non-Abelian, not about the group being non-Abelian.

4. Conserved quantities need not close under multiplication. In closed dynamics, A conserved means $[A, H] = 0$, and the conserved quantities form an algebra. That is an accident of reversibility. For an irreversible channel Φ the conserved quantities are $\mathcal{F} = \{A : \Phi^\dagger(A) = A\}$, a self-adjoint unital *operator system* but in general not an algebra: $A, B \in \mathcal{F}$ does not give $AB \in \mathcal{F}$. It becomes an algebra exactly when Φ has a faithful invariant state (Frigerio [11]). The failure is physical, not a technicality — R22 and R24 turn on it.

5. Six fields have independently found one theorem. Superselection sectors, decoherence-free subspaces, noiseless subsystems, operator-algebra error correction, information-preserving structures, and Krylov/commutant fragmentation are, mathematically, one structure theorem seen from six directions. This is good news — results transfer — and bad news, because each field kept its own notation. §5 lists the traps.

2 A reading path

2.1 If you read only three things

For working in the area of this project, these three suffice; the rest of §2 is reference material.

1. **Wolf**, *Quantum Channels & Operations: Guided Tour* [36]. Free lecture notes, finite-dimensional throughout, and the best single source for the structure theory of channels: Kraus, Stinespring, Choi, the multiplicative domain, fixed points, peripheral spectrum. Written for people who want to *use* it. Start here.
2. **Albert**, *Lindbladians with multiple steady states* [2]. A 134-page thesis that reads as a review, and the most physicist-friendly account of the asymptotic decomposition $\rho(\infty) = \bigoplus_k p_k \rho_k \otimes \sigma_k$ on which R22 and R24 are built. Chapters 1–3 are the relevant ones.
3. **Moudgalya & Motrunich**, *Hilbert Space Fragmentation and Commutant Algebras* [25]. The paper that made “commutant algebra” the standard language for fragmentation, and whose criterion this project’s n_{wcc} literally computes.

Add **Moudgalya, Bernevig & Regnault** [27] if a thesis introduction needs a broad review of fragmentation and scars to cite; it is the standard one and is pedagogical.

2.2 Foundations: what an algebraic formulation is

- **Strocchi** [32] — short, physicist-facing, the gentlest account of *why* one would formulate quantum mechanics algebraically. Read before any mathematics book.
- **Landsman** [19] — open access, comprehensive, with self-contained appendices on functional analysis and C^* -algebras. The best single reference for the foundations proper. Use as a reference, not a cover-to-cover read.
- **Bratteli & Robinson**, vol. 1 [6] — the standard mathematical-physics source. Volume 1 covers C^* - and W^* -algebras, symmetry groups and decomposition of states, which is the relevant part; volume 2 is equilibrium statistical mechanics and is not needed here.
- **Haag** [12] — where the superselection language comes from. Cite for provenance; do not read for technique.

For the pure mathematics, if a proof is ever needed: **Murphy** [28] is the readable introduction, **Blackadar** [4] the encyclopaedia, **Takesaki** [33] the definitive treatment. In finite dimensions — where this entire project lives — none is strictly necessary: the structure theorem is Wedderburn–Artin, and it is linear algebra.

2.3 Quantum channels and open-system structure

- **Wolf** [36] — the primary source, as above.
- **Watrous** [35] — rigorous, finite-dimensional; best for precise statements of Choi and Stinespring, and for anything information-theoretic.
- **Heinosaari & Ziman** [13] — states, effects and channels done carefully; useful if measurement theory is also needed.
- **Breuer & Petruccione** [7], **Rivas & Huelga** [30] — the standard open-systems texts. Use for the physics of Lindblad dynamics; neither emphasises algebraic structure.

- **Baumgartner & Narnhofer** [3] — the structure of the set of stationary states, and the source of the term *enclosure*.
- **Carbone & Pautrat** [9] — the irreducible decomposition done carefully, with the transient/recurrent split explicit. The most mathematically precise of the group.
- **Ticozzi & Viola** [34] — invariance versus *attractivity*. Worth knowing: the gap between an invariant subspace and one actually reached is exactly what R24 flags between its Prop. 3 and its absorption measurements.

2.4 Symmetry in open systems

- **Buča & Prosen** [8] — the strong/weak symmetry distinction. Short, and the definition every later paper uses.
- **Albert & Jiang** [1] — conserved quantities and asymptotic structure, stated for physicists.
- **Lieu *et al.*** [24] — what symmetry breaking means for a Lindbladian, and the observation that a strong-symmetry-broken phase can protect a qubit where a weak one protects only a classical bit. Directly relevant to R22 §8.
- **Lessa *et al.*** [20] — strong-to-weak spontaneous symmetry breaking in mixed states; a third use of “strong/weak” (§5).

2.5 Fragmentation and commutant algebras

- **Sala *et al.*** [31] and **Khemani, Hermele & Nandkishore** [15] — the two papers that introduced fragmentation, independently and simultaneously. Cite both.
- **Moudgalya & Motrunich** [25] — the commutant-algebra formulation. Key definitions: the *bond algebra* generated by the local terms, its *commutant*, and the criterion that fragmentation is exponential growth of the commutant’s dimension.
- **Moudgalya & Motrunich** [26] — symmetries as frustration-free ground states of a local superoperator; a computational route to commutants.
- **Moudgalya, Bernevig & Regnault** [27] — the review.
- **Li, Sala & Pollmann** [21] — fragmentation in *open* systems, with the observation that dephasing reduces quantum fragmentation to classical. The nearest published neighbour to this project.
- **Iadecola** [14] — fragmentation combined with symmetry gives exponentially many logical qubits in degenerate pairs. The cleanest statement of why a non-Abelian commutant is the thing to want.

2.6 Error correction, algebraically

- **Knill & Laflamme** [16] — the correctability conditions.
- **Zanardi & Rasetti** [37], **Lidar, Chuang & Whaley** [22] — decoherence-free subspaces.
- **Knill, Laflamme & Viola** [17] — noiseless *subsystems*; the paper that made move 2 of §1 explicit.

- **Lidar & Whaley [23]** — the review. Read this rather than the primary papers for the picture in one place.
- **Kribs, Laflamme & Poulin [18]** — operator-algebra quantum error correction, unifying DFS, noiseless subsystems and standard codes as special cases of one algebraic statement.
- **Blume-Kohout, Ng, Poulin & Viola [5]** — every information-preserving structure is a matrix algebra. The theorem that lets R22 §8 say the counts *are* code parameters rather than analogies.

3 The three structure theorems

All three are stated in the finite-dimensional form, the only form this project needs.

3.1 Von Neumann’s bicommutant theorem

For $S \subseteq \mathcal{B}(\mathcal{H})$ write $S' = \{A : [A, S] = 0\}$, the *commutant*.

Theorem 1 (Bicommutant). *Let $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H})$ be a $*$ -closed subalgebra containing $\mathbb{K}$. Then $\mathcal{M}'' = \mathcal{M}$ in finite dimensions; in general $\mathcal{M}''$ is the closure of $\mathcal{M}$ in the weak operator topology.*

Physical content: *taking the commutant twice returns you to the start*, so an algebra of observables and its commutant determine each other. This is why one may work with either the symmetry algebra or the dynamics algebra and lose nothing. See [6, 4]; for the finite-dimensional statement, [36, 35].

3.2 The structure theorem (Wedderburn–Artin)

Theorem 2 (Structure of a finite-dimensional $*$ -algebra). *Let $\mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$ be a unital $*$ -closed subalgebra with $\dim \mathcal{H} < \infty$. Then there is a unitary under which*

$$\mathcal{H} \cong \bigoplus_k \mathcal{H}_{n_k} \otimes \mathcal{H}_{m_k}, \quad \mathcal{A} \cong \bigoplus_k M_{n_k} \otimes \mathbb{K}_{m_k}, \quad \mathcal{A}' \cong \bigoplus_k \mathbb{K}_{n_k} \otimes M_{m_k}.$$

Hence $\dim \mathcal{A} = \sum_k n_k^2$ and $\dim \mathcal{A}' = \sum_k m_k^2$.

This is the theorem behind move 2 of §1. The three-way reading of the labels is worth internalising:

symbol	superselection language	code language
k	sector, charge	the classical label
n_k	multiplicity	the protected <i>subsystem</i> (qubits)
m_k	sector dimension	the gauge/syndrome factor (erased)

An algebra is *Abelian* iff every $n_k = 1$, and a *factor* iff there is one k . The centre $\mathcal{A} \cap \mathcal{A}'$ is spanned by the block identities, so $\dim(\mathcal{A} \cap \mathcal{A}')$ counts the sectors. In this project’s terms, the number of blocks is the number of superselection sectors and $\log_2 n_k$ is the number of protected qubits in block k .

3.3 The asymptotic decomposition of a channel

Theorem 3 (Baumgartner–Narnhofer; Albert–Jiang). *Let Φ be a channel on a finite-dimensional $\mathcal{H}$. Then $\mathcal{H} = (\bigoplus_k \mathcal{H}_{n_k} \otimes \mathcal{H}_{m_k}) \oplus \mathcal{H}_{\text{tr}}$, and every state converges under Cesàro averaging to*

$$\rho(\infty) = \bigoplus_k p_k \rho_k \otimes \sigma_k,$$

where $p_k = \text{Tr}(J_k \rho_0)$ is fixed by the initial state, each σ_k is a fixed full-rank state determined by Φ alone, and each ρ_k is an arbitrary state on $\mathcal{H}_{n_k}$ evolving unitarily, $\rho_k \mapsto U_k \rho_k U_k^\dagger$.

Read it as: p_k is a classical label conserved from $t = 0$; ρ_k is quantum memory that survives forever; σ_k is junk. The transient part $\mathcal{H}_{\text{tr}}$ has no closed-system counterpart and is where most of this project’s subtleties live. See [3, 1, 2, 9]; the code-theoretic reading is [5].

4 Glossary

Grouped by theme, alphabetical within each group. Each entry gives a definition, an intuition, and what to cite.

4.1 Algebra

Bond algebra The algebra generated by the local terms of a Hamiltonian or circuit, $\mathcal{A} = \langle h_1, h_2, \dots \rangle$. In the fragmentation literature the dynamics is specified by its bond algebra and the symmetries by its commutant. [25]

Centre $Z(\mathcal{A}) = \mathcal{A} \cap \mathcal{A}'$. By Thm. 2 it is spanned by the block identities, so $\dim Z(\mathcal{A})$ is the number of superselection sectors. “Splitting an algebra by its centre” means finding those blocks. [6]

Commutant $S' = \{A \in \mathcal{B}(\mathcal{H}) : [A, S] = 0 \ \forall S \in S\}$. Always an algebra, even when S is not; a von Neumann algebra when S is $*$ -closed. [6, 25]

Conditional expectation A completely positive unital $E : \mathcal{A} \rightarrow \mathcal{B} \subseteq \mathcal{A}$ with $E|_{\mathcal{B}} = \text{id}$ and $E(bab') = b E(a) b'$ for $b, b' \in \mathcal{B}$. The algebraic version of “trace out” or “project onto the conserved part”. [36, 19]

C^* -algebra A Banach $*$ -algebra with $\|a^*a\| = \|a\|^2$. In finite dimensions, exactly a direct sum of matrix algebras. [28, 19]

Factor A von Neumann algebra with trivial centre; equivalently a single block in Thm. 2. [6]

GNS construction From a state ω on $\mathcal{A}$, builds a Hilbert space, a representation π_ω and a cyclic vector Ω with $\omega(a) = \langle \Omega, \pi_\omega(a) \Omega \rangle$. The bridge from the algebraic formulation back to the familiar one. [6, 32]

Operator system A self-adjoint subspace containing $\mathbb{K}$, closed under linear combination and adjoint but *not* necessarily under products. The correct description of the conserved quantities of an irreversible channel. [36]

Superselection sector A block k in Thm. 2: coherent superpositions across distinct k are unobservable and dynamically irrelevant. [12, 6]

von Neumann algebra A $*$ -subalgebra $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H})$ with $\mathbb{K} \in \mathcal{M}$ and $\mathcal{M} = \mathcal{M}''$. In finite dimensions, any unital $*$ -subalgebra. [6, 4]

4.2 Channels

Choi–Jamiołkowski isomorphism The bijection between maps $\Phi : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$ and operators $J(\Phi) = (\Phi \otimes \text{id})(|\Omega\rangle\langle\Omega|)$; Φ is completely positive iff $J(\Phi) \geq 0$. [36, 35]

Completely positive (CP) $\Phi \otimes \text{id}_n$ is positive for every n . Strictly stronger than positivity, and the correct notion for a physical map. [36]

Dual (adjoint) channel $\Phi^\dagger$, defined by $\text{Tr}(\Phi^\dagger(A)\rho) = \text{Tr}(A\Phi(\rho))$ — the Heisenberg picture. Φ is trace preserving iff $\Phi^\dagger$ is unital. Almost every algebraic statement is made for $\Phi^\dagger$. [36]

Faithful state A state of full rank; equivalently $\omega(a^*a) = 0 \Rightarrow a = 0$. The hypothesis under which $\mathcal{F}$ becomes an algebra. [11]

Fixed-point set $\mathcal{F}(\Phi) = \{A : \Phi(A) = A\}$. An algebra when a faithful invariant state exists, an operator system otherwise. [11, 36]

Irreducible (ergodic) channel One with no non-trivial invariant subspace; equivalently, a unique and faithful stationary state. [9, 36]

Kraus operators $\Phi(\rho) = \sum_j K_j \rho K_j^\dagger$ with $\sum_j K_j^\dagger K_j = \mathbb{K}$. Not unique — different sets are related by an isometry — which matters: a statement should be checked to be representation-independent. [36, 29]

Lindbladian (GKLS generator) The generator of a continuous-time Markovian semigroup, $\dot{\rho} = -i[H, \rho] + \sum_k (L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\})$. [7, 30, 2]

Multiplicative domain For a unital CP map Φ , $\mathcal{M}_\Phi = \{a : \Phi(a^*a) = \Phi(a)^*\Phi(a) \text{ and } \Phi(aa^*) = \Phi(a)\Phi(a)^*\}$: a C^* -algebra on which Φ is multiplicative. Equality in the Kadison–Schwarz inequality is the test. This is why conserved *projections* are better behaved than conserved operators generally. [10, 36]

Peripheral spectrum The eigenvalues of Φ with $|\lambda| = 1$. The corresponding eigenoperators span the asymptotic algebra; the fixed points are only the $\lambda = 1$ part, which is strictly smaller whenever the asymptotic dynamics rotates. [36, 2]

Stinespring dilation Every CP map is $\Phi(\rho) = \text{Tr}_E(V\rho V^\dagger)$ for an isometry V into a larger space: every channel is a unitary on a bigger system. [36, 35]

Unital channel $\Phi(\mathbb{K}) = \mathbb{K}$; equivalently the maximally mixed state is stationary. Unitary evolution is the special case. R22’s Prop. 1 turns on this hypothesis rather than on unitarity. [36]

4.3 Open-system structure

Asymptotic algebra The algebra of peripheral eigenoperators of $\Phi^\dagger$ restricted to the asymptotic support; equal to $\bigoplus_k M_{n_k} \otimes \mathbb{K}_{m_k}$. What the system remembers forever. [2, 5]

Attractivity A subspace is *invariant* if states in it stay, and *attractive* if states outside are drawn to it. These are different, and proving the first does not give the second. [34]

Decoherence-free subspace (DFS) A subspace on which the channel acts unitarily; the special case $m_k = 1$ of Thm. 3. [37, 22, 23]

Enclosure A subspace $\mathcal{H}_0$ with $\text{supp } \rho \subseteq \mathcal{H}_0 \Rightarrow \text{supp } \Phi(\rho) \subseteq \mathcal{H}_0$ — once in, never out. The Schrödinger-picture invariant subspace. [3]

Noiseless subsystem A tensor factor $\mathcal{H}_{n_k}$ inside a block, on which the dynamics is unitary while the co-factor $\mathcal{H}_{m_k}$ is scrambled. Strictly more general than a DFS, and the reason move 2 of §1 matters. [17, 23, 18]

Recurrent / transient A state is transient if the dynamics eventually leaves it with probability one. The recurrent part carries the asymptotics; the transient part is invisible to Thm. 3 and is exactly what makes $\mathcal{F} \neq \mathcal{C}$. [9]

Strong symmetry A unitary U with $[U, K_j] = 0$ for *every* Kraus operator (for a Lindbladian, $[U, H] = [U, L_k] = 0$). Implies a charge measurable at $t = 0$ and a block decomposition of the steady-state space. [8, 1]

Weak symmetry $\Phi(U\rho U^\dagger) = U\Phi(\rho)U^\dagger$: the generator as a whole commutes, but the individual jumps need not. Strictly weaker, and it does *not* give a superselection structure. [8, 24]

4.4 Fragmentation

Commutant algebra (fragmentation sense) The commutant of the bond algebra. **Note the collision:** in the open-system literature “commutant” usually means the commutant of the Kraus set. [25]

Hilbert-space fragmentation (HSF) Exponential-in-system-size growth of the number of dynamically disconnected sectors, not accounted for by any conventional symmetry. Algebraically: exponential growth of the commutant’s dimension. [31, 15, 25, 27]

Krylov sector The orbit of a basis state under the dynamics; the blocks in which fragmentation is usually counted. [31, 27]

Strong vs. weak fragmentation Whether the *largest* sector occupies a vanishing fraction of the Hilbert space (strong) or a finite one (weak). **Unrelated** to strong/weak *symmetry*. [31, 27]

4.5 Error correction

Information-preserving structure Blume-Kohout *et al.*’s classification: anything preserved by a process is a matrix algebra, isometric to the fixed points of an associated unital process. The bridge between Thm. 3 and code parameters. [5]

Knill–Laflamme conditions A code with projector P is correctable for errors $\{E_a\}$ iff $PE_a^\dagger E_b P = c_{ab}P$. Detection of a single error is the same condition with one error. [16, 29]

Operator-algebra QEC (OAQEC) The unification of DFS, noiseless subsystems and standard codes: correctable codes are operator algebras. [18]

Zero-error capacity How much can be recovered with probability exactly one, as against on average. The distinction R22 §3.5 needs: multistability can have ordinary capacity while having none of this. [5]

5 Notational collisions

These have actually caused errors, here or in the literature.

1. **“Strong/weak” means three different things.** (i) Strong/weak *symmetry*: whether each Kraus operator commutes, or only the channel [8]. (ii) Strong/weak *fragmentation*: whether the largest Krylov sector is a vanishing fraction of the space [31]. (iii) Strong-to-weak *spontaneous symmetry breaking* in mixed states [20]. Mutually independent; always say which axis.
2. **“Commutant” has two referents.** Commutant of the bond algebra (fragmentation, [25]) versus commutant of the Kraus set (open systems, [8]). They coincide for closed dynamics and not otherwise.

3. **“Subsystem” has two referents.** A tensor factor of the whole space (physics-degree sense) versus a factor $\mathcal{H}_{n_k}$ of one block (algebraic sense, [17, 18]). Nearly every sentence in the QEC literature means the second.
4. **Conserved quantity $\neq$ symmetry.** $\mathcal{C} \subseteq \mathcal{F}$ always, with equality iff a faithful invariant state exists [11]. Writing “conserved quantity” where “strong symmetry” is meant is a substantive error in open systems, not loose phrasing.
5. **Fixed points $\neq$ asymptotic algebra.** The fixed points of $\Phi^\dagger$ are the $\lambda = 1$ eigenoperators; the asymptotic algebra is the whole peripheral spectrum. They differ whenever the asymptotic dynamics rotates — for a unitary channel, dramatically.

6 Where each piece is used in this project

concept	report	what is computed
commutant algebra, HSF criterion	R21, R22	$n_{\text{wcc}} = \dim(\mathcal{C} \cap \mathcal{D})$
strong symmetry (Buča–Prosen)	R22	weak components of the support graph
enclosure (Baumgartner–Narnhofer)	R22, R24	terminal strong components
conserved quantities $\mathcal{F}$	R22, R24	absorption probabilities; the squeeze
multiplicative domain, projections	R22, R24	every conserved projection is a strong symmetry
Thm. 3 decomposition	R22, R24	$\bigoplus_k M_{n_k} \otimes \sigma_{m_k}$, exactly
noiseless subsystem, DFS	R23, R24	$m_k = 1$ for all eight rules
Knill–Laflamme, code distance	R24	distance 1; no subcode helps
information-preserving structure	R22	bits vs. qubits vs. nothing

Two results of this project are statements *about* the framework rather than applications of it, and are the natural places for a thesis to claim novelty. First, R22’s theorem that the projections in $\mathcal{F}$ are exactly the indicators of unions of weak components, so the Boolean algebra of always-answerable questions cannot grow with the number of attractors. Second, R24’s correction that $\dim(\mathcal{F} \cap \mathcal{D}) = n_{\text{rec}}$ holds only for the monitored chain, the coherent statement being the squeeze $n_{\text{wcc}} \leq \dim(\mathcal{F} \cap \mathcal{D}) \leq n_{\text{rec}}$, with W_{29} at $N = 4$ as an explicit counterexample to equality.

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